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A secure and trustworthy blockchain-assisted edge computing architecture for industrial internet of things

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The Industrial Internet of Things (IIoT) seeks to improve smart factory productivity by leveraging automation and scalability. For automation in industry, optimization, collaboration and protection, and scalability, the IoTs paradigm, technology for communication and information, and intelligent systems are integrated as a single organism. This article presents a blockchain-assisted safe data-sharing mechanism that provides security requirements in the industry using IoT. End-to-end authentication is developed based on the blockchain's reputation, and the smart contract is used to validate nodes' security measures. Through integrity verification and categorization in node terminals and industry, the blockchain paradigm manages data collection and dissemination. An efficient proof of authentication (PoAh) consensus mechanism is created using the blockchain network to build a collaborative network to preserve logs and verification data in the industrial IoT. It accomplishes trustworthy authentication and endpoint activity tracing, and edge computing is used in blockchain nodes to offer device authentication processes that utilize smart contracts and PoAh. According to an experimental study, the proposed architecture lowers the authentication time and obtains a high response rate. The proposed system's service time shows the efficiency of PoAh-based blockchain architecture for industrial IoT compared with existing works. Finally, we evaluated various block sizes to ensure an efficient transaction rate. The outcomes demonstrate the practicality of the suggested and put-into-practice architecture, distinguished by enhanced data audibility, device data ownership, security, and privacy while utilizing decentralized storage.

Keywords Authentication, Blockchain, Edge computing, Industrial internet of things, PoAh

The adoption of industrial facilities using information and communication systems to support dependable production has given rise to the Industrial Internet of Things (IIoT). With the aid of IoT and intelligent computing, automation, self-sufficient processing, and computing capabilities are implemented for smart industries[1]. Through the management layer, the fieldprimary blockchain network implements a layer carries out many duties that support production and customer replies in traditional industry operations[2]. Industries are now equipped to meet their users' growing demands and productivity requirements, thanks to IIoT. Recent developments in robotic and autonomous machines take the place of manual or human-intervened procedures. The potential of machine-oriented jobs to aggregate, analyze and enhance data handling practices and customer replies.

Utilizing pervasive systems, ubiquitous access, distribution of resources, distributing automation, and self-dependent computation are made possible. Infrastructure and applications are made available as amenities in the cloud ecosystem. Data visualization tools, mathematical frameworks, people, applications, intelligent

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computing machines, and IoT can all work together in the cloud. With the help of many automated systems and communication methods, such a diversified platform can improve customer service. The IoT has enormous potential to enhance the features and capabilities of any service it provides or supports, regardless of the use case. An additional dimension of difficulties will arise because of current use cases and near future ones. Centralized data storage is a big obstacle. The existing storage mechanism in IoT system implementations, the quantity of data gathered in one place, and the number of IoT devices all go hand in hand. Concerns about data ownership and administration will arise because of this central data consolidation. Attackers may find centralized data aggregation appealing and manipulate the IoT system to their advantage, leading to catastrophic consequences[3].

Within the IoT environment, this raises confidentiality and security issues. A system that can use the current edge computing platform's features and address some of these problems is needed to facilitate the scaled interconnected future. IoT systems must interact and integrate with their surroundings safely and effectively[4]. These IoTs technologies provide enormous amounts of data that can be instantly processed and analyzed to aid decision-making process. Such processing should occur near the network's edge to maximize timeliness and minimize bandwidth needs. Edge computing can interpret data and make choices at the network's edge. As a result, it is necessary to investigate the advantages and disadvantages of edge computing in IoT.

Edge computing: an overview

The massive influx of data from the IoT is straining cloud computing due to expensive bandwidth and high latency. However, to meet the fundamental needs of the industry in areas such as real-time enterprise, artificial intelligence, security, and privacy, edge computing offers a service computing model that is near objects or data sources, enabling rapid response to various services[5]. In this section, we will outline the key features of edge computing. With edge computing, processing power is physically and conceptually near to the endpoints. This means that everything from data production to data processing to data utilization occurs near the data source, ensuring a near-instantaneous response to requests from terminals.

Edge servers can self-recover and operate independently in a network outage. The central cloud is responsible for executing flexible computational offloading and assigning tasks to appropriate edge servers. Many diverse and adaptable edge devices are available to meet the increasing demands of the IoTs. In addition, servers located in edge computing surroundings may enhance cloud computing capabilities by providing these nearby diverse edge devices with enhanced storage, processing, and communication features. These enhanced features include increased data storage capacity, faster data processing speeds, and improved communication bandwidth, all of which contribute to the efficiency and performance of edge computing. This shifts resources away from the cloud facilities and onto the edge, reducing the load on the cloud layer. Restricted Data Access is unnecessary to transport data to distant clouds since edge devices can collect and analyze data locally. Thus, security is improved to some level since most information, particularly private data, does not have to pass over the network.

With increasing processing power and storage space, the architecture progresses from the device layer to the core layer and back again. The device layer comprises many diverse IoT devices with limited resources and computing capability[6]. These devices include sensors, RFID tags, recording devices, automobiles, outside units, etc. Their primary purpose is to gather, send out, and upload raw data, serving as the data collection and transmission hub of the IoT system. The edge layer supplies processing, storage, and network bandwidth to devices at the network's periphery, acting as servers from a computational standpoint. The computer offloading methodologies of jobs and the coordination of associated resources are crucial to the edge layer because of the nature of edge server capacities and network architecture. The core infrastructure typically indicates the cloud layer, whose topologies are comparable to cloud computing. The cloud layer must be invoked to finish the jobs the edge layer cannot handle. In addition, the cloud layer can adapt the edge layer's deployment techniques in real time according to the constantly changing allocation of edge resources.

Blockchain: an overview

Everyone has been talking about blockchain technology ever since Bitcoin was developed. A blockchain is a distributed ledger technology that can record financial transactions[7]. A hash label connects the blockchain's most recent block to the one that came before it. In particular, the following data can be stored in a blockchain block:

1. Information on the activities, including the time, date, and amount.
2. Facts regarding the participants in those transactions.
3. A distinct hash code that distinguishes one block from another.

A new block is appended to the end of the blockchain for each transaction. Since the blockchain stores all transactions in a public ledger, which is a type of blockchain that is open to everyone, it is entirely transparent. Users' identities and other sensitive information will be maintained securely on the public blockchain to protect their privacy[8]. In addition to the hash of the data inside it, each block on the blockchain also includes the hash of the block that came before it. Therefore, a decentralized distributed ledger technology (DLT) is a common way to describe blockchain. It is challenging, if possible, for a hacker to alter a blockchain transaction without changing the hash of every block that follows it. Blockchain is well-suited for use in several industries because of its inherent security, which includes finance, healthcare, government agencies, managing supply chains, and many more[9].

Blockchain transactions are confirmed and verified by a "consensus protocol," as opposed to conventional systems that rely on a central authority to do the same. Using a consensus method, every node in the blockchain network may agree on the blockchain's present state[10]. A consensus algorithm must be implemented to ensure that all blockchain nodes agree on the present condition of the chain every time a new block is produced (via

transactions). The consensus process is also performed when a new node is added to the blockchain. Therefore, consensus algorithms play a crucial role in establishing the trustworthiness of every node in the blockchain, ensuring the blockchain's dependability and security. Proof of Capacity, Proof of Stake, Proof of Burn, and Practical Byzantine Fault Tolerance are among today's most widely utilized consensus algorithms. Smart contracts are implemented on a blockchain to guarantee that the transactions adhere to the specified conditions and limitations.

Smart contracts, implemented with just a few lines of code, ensure that each transaction adheres to pre-agreed conditions. This efficient automation relieves you from the manual task of confirming transactions, empowering you to make rapid and accurate decisions. The decentralization, immutability, and transparency of blockchain further enhance its appeal.

(1) Decentralization: Before blockchain technology became popular, there used to be a single location where all the data was kept, and everyone could access it through this one location. One point of failure, attack susceptibility, and other problems characterize centralized systems. Because all the nodes in a decentralized system have the data, it is possible to circumvent these problems with centralized systems.
(2) Immutability: Data recorded on a blockchain cannot be altered because of consensus algorithms. Blockchain is perfect for use in many industries because to this quality, including banking, supply chain management, regulation etc[11].
(3) Transparency: Each blockchain technology is perpetually open source. Anybody sees every transaction on the blockchain. Since changes to the blockchain network require approval from most nodes, the technology or transactions remain safe despite their transparency. Complex cryptography techniques are used to conceal user information[12].

Implementing blockchain technology in decentralized systems, such as IoT, presents several significant challenges despite its numerous advantages. Running consensus procedures, like PoW, on the blockchain causes considerable network latency and substantial energy consumption, a severe issue. For dispersed IoT networks where IoT devices have limited resources, this can be a dealbreaker for blockchain applications. The low data transfer rates of blockchain platforms is another issue.

Trust requirements for edge-based IoT systems

Administration of trust for edge-based in the pursuit of a dependable system connecting IoT nodes to the edge network, IoT becomes an essential subject. The security challenges of edge computing-based IoT systems are shown in Fig. 1.

- *Precision* It indicates how near a node's computed trust value is to its fundamental or trust value when all information about its behavior is known. Accuracy in trust is crucial for the trust system of management[13].
- *Security* If you want to be sure your edge-based IoT system is secure, you need to build trust. When it comes to the infrastructure layer of edge servers, it does offer methods to guarantee security.
- *Accessibility* It means that the trust management system can guarantee that all network services will be fully available regardless of resources are attacked or unavailable for any other reason; this is especially frequent in edge networks.

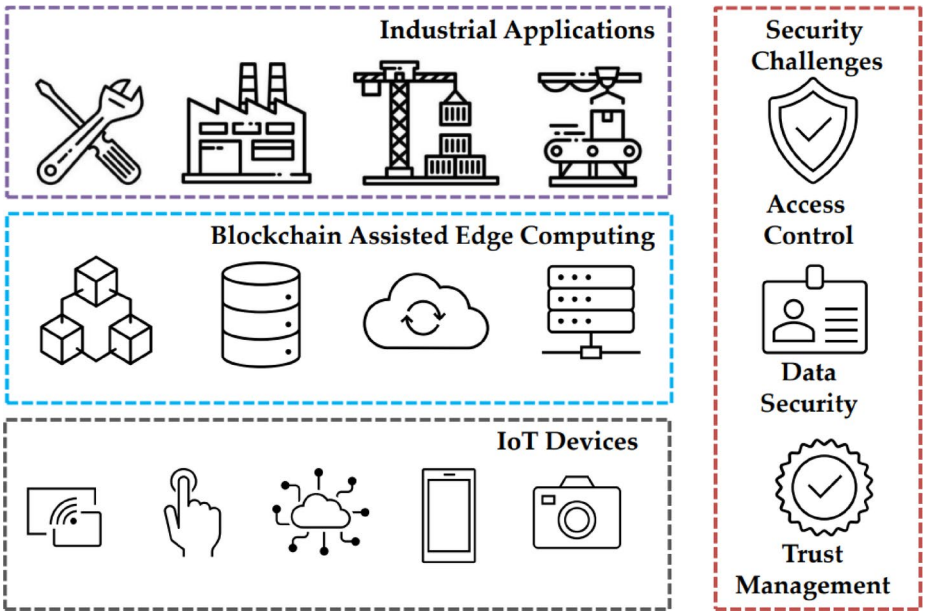

Fig. 1. Edge computing based IoT systems and its security challenges.

- *Dependability* Trust management approaches rely on it as a crucial parameter. The lifespan of the edge-based network guarantees that all functionalities perform correctly.
- *Diversity* Typical components of edge-based IoT include diverse subnets, nodes, edge nodes, and edge servers. For a trust management system to meet the diversity criteria, it must be able to accommodate objects with varying degrees of complexity, computational capacity, and use of energy.

- *Lightweight* Restricted devices and networks are a part of IoT systems that operate at the edge. Because of this, edge-based IoT trust management frameworks need to be small and efficient enough to function effectively across a wide range of IoT nodes, networks, and servers, even when faced with limited resources[14].
- *Scalability and flexibility* The IoT systems are constantly evolving, edge-based IoT must be very flexible and peers participating in trust-based transactions can change quickly. Additionally, regulations for assessing the credibility of edge servers and nodes are subject to frequent device changes, and the availability of their resources might vary widely[15]. A new node can be introduced to the system, and a different application may need connectivity to these edge servers and nodes; nevertheless, an IoT device's transmission and storage costs do not scale with the number of devices in the network.

Motivations and contributions

Blockchain and IIoT systems are inherently different, posing a significant challenge in their integration with edge computing. The limited resources of IoT devices and the need for a scalable security solution in the face of extensive IIoT networks further complicate industry system. However, this study proposes a private blockchain-based architecture for the IIoT network that is lightweight, scalable, and specifically designed to tackle these difficulties. The potential of this proposed design is immense, offering a ray of hope in addressing the challenges of IIoT integration with blockchain. The following are some of the most important contributions of the proposed system.

1. Integrating blockchain with IoTs in a way that works for smart factories is a significant opportunity. This study proposes a private, decentralized, blockchain-based IIoT network that is not only lightweight but also highly extensible, ensuring its adaptability to the evolving needs of the industry.
2. In the proposed architecture, first step is implementing small nodes that use virtual machines to conduct asymmetric cryptography in real time. Secondly, the primary blockchain network implements a (PoAh) system. This system, which is both scalable and compatible with the IoT, ensures that only authenticated devices can participate in the blockchain network, thereby enhancing its security and reliability.
3. Further, we examine the blockchain integrated IIoT paradigm's security needs. We specifically examine the way blockchain might help with important industry security services such as trust management, secure data sharing, device registration, and device authentication.

The outline of the rest of the article is as follows: Sect. 2 covers the current state of blockchain technology as it pertains to IIoT systems. Section 3 details the edge computing-enabled blockchain architecture. Section 4 details the proposed architecture and its operation model. Section 5 provides a comprehensive analysis of the system's efficiency. Lastly, Sect. 6 provides a concise summary[16].

Related work

The (IIoT) has emerged as a game-changing technology in the past decade, drawing heavy interest from academics looking to improve manufacturing operations[17]. Several industries have seen the benefits of the IIoT in areas such as environmentally friendly procedures, proactive upkeep, precise real-time data processing, and more. When used in industrial settings, blockchain technology may ensure human rights are respected while providing traceability and openness. In this section, we summarize several recent studies that have addressed blockchain's potential uses in IIoT. Minoli, D., and Occhiogrosso[18] state that IoT devices may benefit from blockchain for legitimate transactions and untampered archives in their research using blockchain as an IoT platform. According to them, the most challenging part of implementing blockchain technology in IoT devices is finding a suitable hosting infrastructure. Testing both fog and the cloud as potential implementation platforms, they find that fog is superior. In their work on edge computing for blockchain in endpoints with restricted resources, Feng et al.[19] introduced a prototype. They proved that mobile devices may access and use edge network computing power to enable blockchain in their apps. As more miners sign up for the network, the integration also benefits service providers, according to their testbed and experimental findings.

To help video streaming systems employ blockchain for transcoding, Liu et al.[20] suggested a methodology for distributing resources in edge computing. They devised a system to incentivize transcoding and looked at two ways to remove the workload from edge computing. Their simulation findings confirmed the model's ability to maximize average profits in offloading and dynamic allocation of resource schemes. To determine if it would be possible to use blockchain technology and peer-to-peer (P2P) communication to create a decentralized private design for IoT applications, Alzoubi et al. conducted a systematic literature study on the topic[21]. Their research revealed that three problems anonymity, adaptability, and integrity impeded smooth integration. Their research showed that identities are the exclusive means of guaranteeing anonymity, that integrity depends on PoW's difficulty and the substantial number of honest miners, and that flexibility is, by coincidence, constrained by PoW's complexity.

Security and accountability of recorded resource administration status data are critical components of an effective resource management system. Integrating blockchain technology into such architecture may guarantee data security and provide a reliable environment. To achieve the methodically integrated management of resources and the decentralized construction of ecosystems during the creation and activities (DevOps) process,

a blockchain-based management architecture is suggested in[22]. The structure would use smart contracts and blockchain technology to offer trustworthy services independent of any third party. Additionally[23], uses blockchain technology to record the hash values of sensitive data and provide tracking services, ensuring accuracy and accountability.

Edge computing can also benefit from blockchain smart contracts, increasing its security. To improve the reliability of collaborative edge computing (CEC), the authors of[24] introduce BlockEdge, an application framework driven by blockchain technology. Based on the on-chain validation findings and deposit balances, BlockEdge builds a trust rating system. To address the need for a trust assessment method and to maintain privacy in the context of trust management and data security, a new blockchain called TrustChain is suggested[25]. A blockchain-based trust process for transferring task analyzing in mobile edge computing (MEC) is offered in[26]. This mechanism aims to avoid incidents of self-centered edge servers and counterfeit service records by deploying a blockchain to store and handle trust data. Additionally, it develops a consensus protocol that combines PoW and PoS.

The methods for allocating CPUs and evaluating trust on edge servers are not just individual implementations, but collaborative efforts that draw from two different schools of thought-RL and DRL. Both these approaches are harnessed to optimize the number of CPUs on edge servers, thereby reducing their computational consumption and evaluating their service reliability using Bayesian reasoning. By employing convolutional neural networks (CNNs) to reduce the current state space and incorporating an altered Boltzmann distribution to achieve the appropriate number of CPUs, the latter method promises to further enhance computational speed. The steps for controlling access among devices and gateway nodes, gateway nodes and servers at the edge, and cloud servers are all part of a consortium blockchain-enabled access control initiative in edge computing-based generic IoT environments, as suggested in[27]. This collaborative effort is set to redefine access control in edge computing. To fortify data against attacks such as identity-revealing, session hijacking, relay, and man-in-the-middle, a decentralized security system has been developed in[28] using blockchain technology and edge computing. This is a testament to the power of collaboration in the face of common threats.

Zhang et al.[29] presented the application of industrial IoT for effective certificate-less signing in cloud-assisted systems. For data reliability and third-party security, lightweight certificate-less verification is ideal. Reliable certificateless signatures (RCLS) prevent signature forgeries. A secure sharing mechanism was proposed by Zheng and Cai[30] to ensure the privacy of IIoT data, which is significant in the context of increasing concerns about data privacy and security in IIoT systems. The main features of this system are data sharing upon request and profit-based differentiated privacy. From confidentiality of information to bandwidth utilization, this framework excels at IoT security considerations. A credit-based consensus method is the basis for Secure IIoT. An IoT device PoW system based on credits was discussed by Huang et al.[31]. The IoT device sensitive information is safeguarded by blockchain ledger.

Zhu et al.[32] discussed an IIoT-directed network for secure identification. The third-party and the authorized user have a secure connection. The encryption accumulator is an element of the direct graph that facilitates this process. An adaptable signature system is employed to ensure the safety of data trade. Using blockchain technology and federated learning, Lu et al.[33] suggested a method for exchanging data in IIoT that preserves users' privacy. This method involves a combination of blockchain technology and federated learning to ensure data privacy and security in IoT systems. In maintained federated learning, data exchange is implemented so that different people can engage with the computer. To train their datasets, privacy-preserving and federated approaches both purchase blockchain technology. A deep belief network–based supervisory control and data collecting system was introduced by Huda et al.[34]. The writer devised ensemble techniques for deep belief learning and categorization. System control is accomplished using the industrial control system. Both the traffic and payload features are handled.

The IIoT uses a private data protector to prevent confidential data leakage. To keep an eye on data and adjust the distributed defense assault, Wu et al.[35] suggested a hierarchical analytical procedure and approach for ordering preferences based on their similarity to an ideal answer. The use of the Ethereum blockchain and deep reinforcement learning in tandem[36] for gathering and exchanging data from industrial IoT devices is another key aspect. However, it's the consideration of sensor distance and power sources that truly aids in obtaining high-quality data, a crucial piece of information that our audience will find enlightening. This is accomplished using intelligent mobile terminals, efficiently exchanging data and gathering trustworthy information. Most of the data may be obtained by employing DRL. The dataset is allocated to ensure the input contributors' security using federated deep learning. The lightweight technique's multi-contributors are guaranteed vital privacy preservation.

Gai et al.[37] built IIoT based on a blockchain that uses asymmetrical privacy. This piece aims to bridge the gap between blockchain IoT and BIoE. The IoT may create a privacy-indicating energy cost indicator using its controls. Data is collected in a way that doesn't impact power consumption. Zhou et al.[38] suggested a public key encrypted text for industrial IoT devices to use in lightweight, safe searching. Edge devices, which are computing devices that operate at the edge of the network, are used to enable cryptographic activities in communication between edge devices. The goal is to safeguard the edge device as it makes quick computations.

The rising significance of safe and effective data processing systems to handle the massive amounts of data produced by industrial applications is highlighted in the above literature. Because edge computing can process data closer to the source, reducing latency and bandwidth usage has become an essential component. Traditional edge and cloud-based approaches frequently need more security and confidence than blockchain's decentralized, immutable ledger provides. Related work has investigated several frameworks that combine edge computing and blockchain to manage identities, encrypt data, and preserve integrity among dispersed devices in IIoT networks. However, the adoption of many existing models in IIoT systems with limited resources could be improved by scalability, energy consumption, and interoperability issues. Limitations still exist despite these developments,

especially when it comes to effectively striking a balance between security and performance in blockchain-based computing frameworks. Low-power IoT devices may need help to handle the high processing demands of blockchain consensus methods, which plague many current solutions. Furthermore, creating trust models that effectively manage dynamic device involvement and offer smooth interoperability across various IIoT platforms presents difficulties. The proposed system is employed to develop blockchain-based edge computing architectures suited to industrial IoT applications' requirements.

Lightweight blockchain for resource-limited edge devices

Blockchain demands a lot of processing power for consensus techniques and sufficient space to store the whole blockchain when it allocates resources in edge computing. Therefore, the most critical problem in blockchain-based resource allocation is enabling blockchain expansion in edge computing. In[39], researchers have looked at several ways to improve lightweight blockchains. An efficient LiTiChain blockchain is suggested in[40] to accommodate the edge-enabled IoT system. LiTiChain uses an End-time Ordering Graph (EOG), a tree-like data structure in blockchain technology, to keep track of time-sensitive activities and blocks. To save storage space on the edge nodes, the EOG depends on the sequence of the end time and deletes expired blocks. The design of EOG is adjusted to preserve the order of blocks using algorithms for k-height inclusion and deletion/renewal of a block.

Our proposed blockchain, tailored for edge computing, is not just a theoretical concept. It is a practical solution that ensures fair storage and speedy, effective use of blocks. The article presents solutions for catching recent blocks and the best distribution of storage space for blocks[41]. In addition, nodes are incentivized to own more tokens and retain more data using an enhanced PoS consensus process tailored for edge devices with limited battery life. To overcome the mathematical mystery of PoW consensus, a successful consensus technique is suggested in[42] that uses the expectation-maximization algorithm with polynomial matrix factorization as its basis.

Edge computing-enabled blockchain

Edge devices may support blockchain with minimal resources thanks to edge computing, a new paradigm. Put differently, edge computing is a facilitator of blockchain resources. A MEC-enabled blockchain architecture for IoT is presented in[43], wherein cloud computing resources may be purchased by IoT devices using mining as a service. In[44] MEC servers are separated into authentication servers (AS) and computing servers (CS) to reduce computing and transmission costs for vehicles with limited resources. Batch confirmation and anonymity techniques improve privacy and authentication effectiveness. The plan mostly depends on MEC servers' dependability and accessibility, which might be problematic in areas with inadequate infrastructure or erratic connectivity.

Feng et al.[45] introduces PBAG, a blockchain-based authentication mechanism for the Internet of Vehicles (IoVs) that protects privacy. IoVs allow for real-time communication between infrastructure and automobiles, but they also present privacy concerns like monitoring and security risks like message forging. It uses a dynamic updating method supported by blockchain technology to guarantee certificate validity instead of relying on laborious Certificate Revocation Lists (CRLs). Due to its heavy reliance on trusted authority and blockchain infrastructure, the protocol may be difficult to implement in rural or impoverished areas. It offers defense against common assaults and has been tested through simulations, demonstrating its effectiveness in high-demand IoV scenarios. It can be challenging to balance traceability and privacy, and if the traceability mechanism is broken, unexpected dangers could occur. In[46], two mining strategies are examined to maximize earnings for edge service suppliers and miners: immediate reporting following practical computing and selective reporting following successful computing. To increase the adaptability of blockchain, an architecture for edge computing-based parallel computation has been developed[47]. The blockchain, acting as a decentralized database, plays a crucial role in synchronizing and preserving the activity information of IoT gadgets, thereby ensuring data integrity and security. Edge nodes are set up to carry out the parallel analyzing method by providing the verification service.

The research has provided valuable insights into the practicality of combining blockchain technology with edge computing for IoT applications. This knowledge, derived from studies that have used integration as a launchpad for applications, delves into the broad architectural challenges of integration and the intricate world of mitigating specific blockchain concerns. Our research has focused on developing a practical platform for implementing blockchain services in the IoT, with specific emphasis on meeting the recognized critical security and flexibility requirements for safe, decentralized storage.

The proposed blockchain enabled edge computing architecture for IIoT

The proposed structure aims to make safe handling of operations possible in intelligent industrial settings. These functions include initializing the system, storing data from sensors and actuators, and providing reliable industrial operations to clients. The following subsections give a brief description of each of these operations.

System architecture

We present a high-speed blockchain-enabled edge computing architecture tailored for medical applications. The proposed architecture, as depicted in Fig. 2, offers significant benefits. IoT users can efficiently dispatch tasks to the industry through a unified interface. The blockchain structures, which specify data dependencies, streamline the organization of industry tasks. The manager, leveraging the blockchain structure of operations, resource utilization, and network circumstances of edge servers, devises a coordination strategy.

Simultaneously, the manager leverages the blockchain to ensure the security and authenticity of user identification. If a user's actions are deemed unauthorized, the manager can immediately halt the allocation of

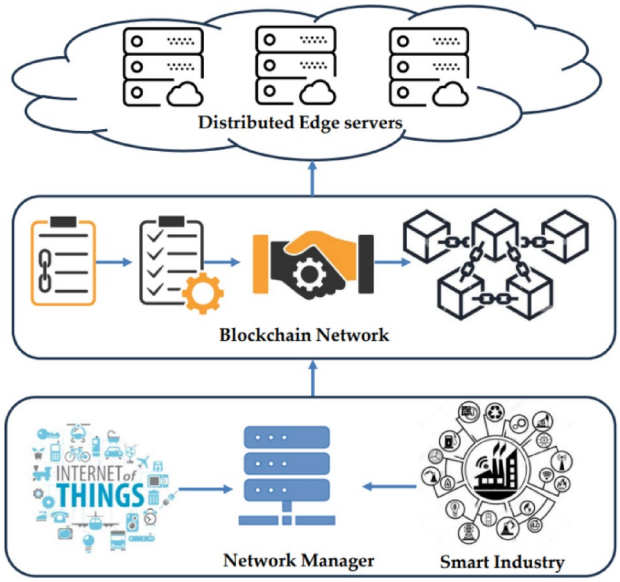

Fig. 2. The proposed architecture for blockchain assisted edge computing based IoT systems.

computer resources. The edge servers, in turn, execute the tasks, download and install the necessary container images, and report the outcomes to the management. Each user is affiliated with an authorized network, which responds to computed models. The job computation results, and user-identifying data are securely stored on the blockchain, ensuring the highest level of security and protection.

All approved user identification data is managed by the gateway and uploaded to the blockchain, which is continuously updated. The blockchain's data is readily accessible upon receiving a query response from the management. In this setup, users can enjoy the benefits of minimal latency, straightforward implementation, and control over an edge computing service. The blockchain, with its robust security and reputation protection, ensures the service's integrity. User clients and associated IoT devices seamlessly register in the blockchain system during the setup phase. This process is crucial for enabling user and device re-authentication. User and device registration are the final stages of the registration process, which is designed to be user-friendly and straightforward.

Main components

1) The manager is responsible for collecting data, task separation, subtask coordination, interim outcome conservation, user authentication, and resource handling. In contrast to conventional decentralized edge computing designs, the proposed architecture securely encrypts the system using blockchain technology and controls the entire system worldwide.

2) The processes are distributed across edge servers. They listen to the work demands and provide the management with resource data. Upon receiving user tasks, it will obtain and launch the necessary container pictures and establish ports for interaction. The manager will arrange for edge servers to process the job and provide the results after receiving the raw data. Since the edge servers are arranged as a changing system, they may be quickly added to and deleted from the platform. As a result, edge computing platforms' expansion and reliability are assured.

3) Industry: Every user has an associated IIoT. In response to the relevant user, the industry will act according to the directions given by the manager. The edge servers' calculations are used to determine the instructions. Specific data are synced to the blockchain for investigation or additional authentication.

4) First, a system administrator creates a new, distinct ID for each client. The admin then submits this ID as an activity request to the blockchain nodes. Using a smart contract, the blockchain node verifies the ID's presence in the system. If the ID is not in use, the transaction is permitted. A new block is created and disseminated to all the blockchain nodes. The blockchain uses its private key, a crucial security measure, to generate a user ID certificate. This certificate is given to the administrator if the user has properly registered. Administrators can use their private keys to access certificate data.

5) Sensors and actuators are found in the suggested design's physical layer. The data from the sensor or actuator is preprocessed as needed, and then the virtual machine, acting as a data handler, forwards it to a tiny node in the blockchain application layer. The portable node processes the device registration demand and sends an extra transaction request to the blockchain node. In the blockchain network, an intelligent contract confirms the device ID. A fresh block is created and dispersed across all nodes on the blockchain after the PoAh procedure is finished.

Blockchain integration

The blockchain serves as the decentralized verifier, ensuring the security and reliability of the system. It stores and validates transactions produced by decentralized entities. The hash values of user-submitted data and the outcomes calculated by edge nodes form a transaction's information structure. Each transaction contains digital signatures and IDs of the relevant edge servers and medical devices, along with a timestamp. The four smart contracts document and validate the data analysis procedure, further enhancing the system's security. The Identity contract plays a crucial role in the system, overseeing all entities, including edge servers and dispersed IIoT users. Its oversight provides a sense of control and management, ensuring the system's smooth operation.

Submission contract This contract serves as an interface, enabling edge servers and users to securely contribute their personal information to the blockchain, ensuring the integrity and privacy of the data.

Verification contract This contract is responsible for overseeing and validating the transactions, ensuring the accuracy and reliability of the data stored in the blockchain.

Credit contract This oversees awarding and disciplining the players. Information collection and blockchain interaction are primarily handled via identification contracts, contribution contracts, and reputation contracts. We can refer to the execution of the contracts as existent solutions. The crucial agreement in the system we use to guarantee data transfer safety is the validation contract included in the blockchain architecture. We illustrate the way a verification contract is implemented as follows.

(1) A sequence of round random numbers is produced decentralized.
(2) Using the produced random numbers, a certain number of nodes will be chosen to participate in the collaboration.
(3) Consortium members will vote to decide if the transaction is legitimate. After the edge servers have finished processing the jobs, the manager will invoke the upload contract to post the data to the blockchain. To verify the identity of the edge servers and IoT users, their submission agreement will initially contact the person who signed the contract. After that, the data will be uploaded as a transaction to the distributed ledger and regularly checked. The monitoring connection, which oversees verifying transactions, will be chosen by the system regularly to increase system efficiency. The consortium participants will vote on the deals. A transaction will be considered legitimate if it obtains the most votes. If not, it will be deemed void, and the relevant parties will be held accountable through a credit contract. In this instance, the system's security is assured. The ability of the blockchain system to facilitate large-scale, real-time concurrency in the healthcare setting presents a particular difficulty.

Consensus model: proof of authentication

This study proposes a novel consensus mechanism, called Proof-of-Authentication, to create a distributed ledger that is durable and compact for endpoints with limited resources. During block confirmation, this consensus technique adds a verification method[48].

The network precipitants create transactions (Trx) at the start of the process by assembling sensed or collected data into blocks. Blocks are shown in the illustration as B = {Trx1, Trx2, …, Trxn}. The nodes broadcast the blocks so that other reliable nodes in the network may review and validate them. We adhere to the typical IoT deployment procedures to establish the initial trustworthy node-set based on geographic location. You can connect to trusted nodes from anywhere in the network. The ElGamal cryptosystem is used by the suggested model for encryption and decryption., i.e.

$$y = g^x \ (mod\ p)$$

Where y denotes the public key PuK, and x denotes the private key PrK. Every network device knows the big prime numbers for generator function g and modulus operation p. The network user signs the block using its private key PrK, or x and makes the public key PuK, or y, available to the network before the block transactions.

The suggested protocol is specially made for dispersed networks with limited resources, like the Internet of Things, where a small number of trustworthy nodes with a minimal trust value will be initialized during network deployment, and other network devices will have a zero-trust value or "tr = 0." The trust value of an authenticated node is increased by one upon successful block authentication. Likewise, every forged block authentication will cause the trust value to drop by 'tr = 1'. Other untrusted nodes in the network can also recognize the authenticated block to obtain trust value, or "tr = 0.5," after node authentication by the trustworthy node. Crucially, detecting fraudulent block authentication might result in a high trust score, or "tr = 1". When a trusted node's trust value falls below the threshold "tr < th," it may be eliminated from the process through this trusted assessment procedure, and a regular node can continue to the authentication process. Here, trustworthy nodes are given an initial trust value of "tr = 10" and a threshold of "th = 5." The authentication process is depicted in Fig. 3.

When the trustworthy node receives the block for validation, it looks for the source public key, or y, to verify the signature. The public key is utilized in this work's asymmetric cryptography to validate signatures. When everything else is publicly accessible, an attacker cannot obtain the private key or x. The discrete log problem is this one. Following successfully verifying the signature, the trustworthy node assesses the Media Access Control (MAC) address and does a second assessment by comparing it with the received one. The trustworthy nodes with PoAh identity broadcast confirmed blocks following successful authentication. Then, individual network users confirm the PoAh data to add blocks to the chain. Following accepting a valid block, the user gets the hash value from the previous block to save into the current block and computes a hash value to connect to the next block.

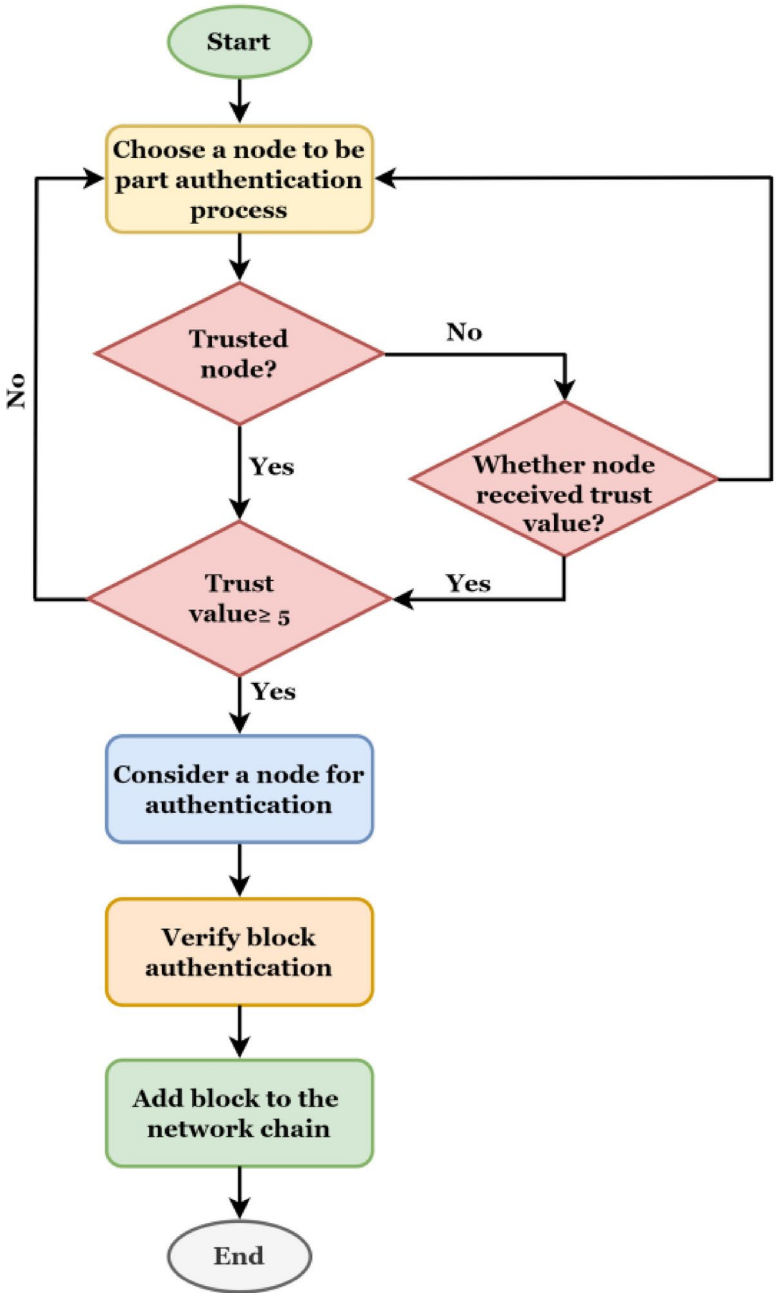

Fig. 3. The PoAh smart contract flow chat.

Simulation results and analysis

The proposed system, evaluated using MATLAB and Hyperledger Fabric, is not only efficient and reliable but also highly cost-effective. The system program functionality is contained in a smart contract, executed by Hyper-Ledger Fabric using the Docker containers technique. This article's testing setup is run in an Ubuntu 16.04 Hyperledger Fabric, version 1.4 Docker 18.06 container, ensuring the system's cost-effectiveness. As coalition peers, multiple nodes are hosted on one server computer, collaborating using the PoAh consensus mechanism to achieve agreements, further enhancing the system's cost-effectiveness. Every node has 8-vCPUS at 2.0 GHz, contributing to the system's high performance and cost-effectiveness. Virtual LAN cards with a speed of 1 Mbps are used to link VMs together, ensuring cost-effective communication. A single ordering service called a typical order is deployed, utilizing the suggested smart contract. Table 1 displays the expenses associated with calculation and communication throughout the authentication process, reaffirming the system's cost-effectiveness.

Figure 4 illustrates that authentication times grow for varying numbers of peers deployed as terminal counts rise from five to fifty. When terminal quantities increase, edge nodes must wait longer to process their authentication queries. The proposed model, with its efficient authentication time and validation period, is

Process module	Cost
Generation of random values (16 bits)	0.45 ms
Hash function computation (160 bits)	2.75 ms
Cryptography paring (176 bits)	9.56 ms
Node verification	0.92 ms
Deployment of PoAh	8.25 ms

Table 1. IIoT device communication and blockchain costs.

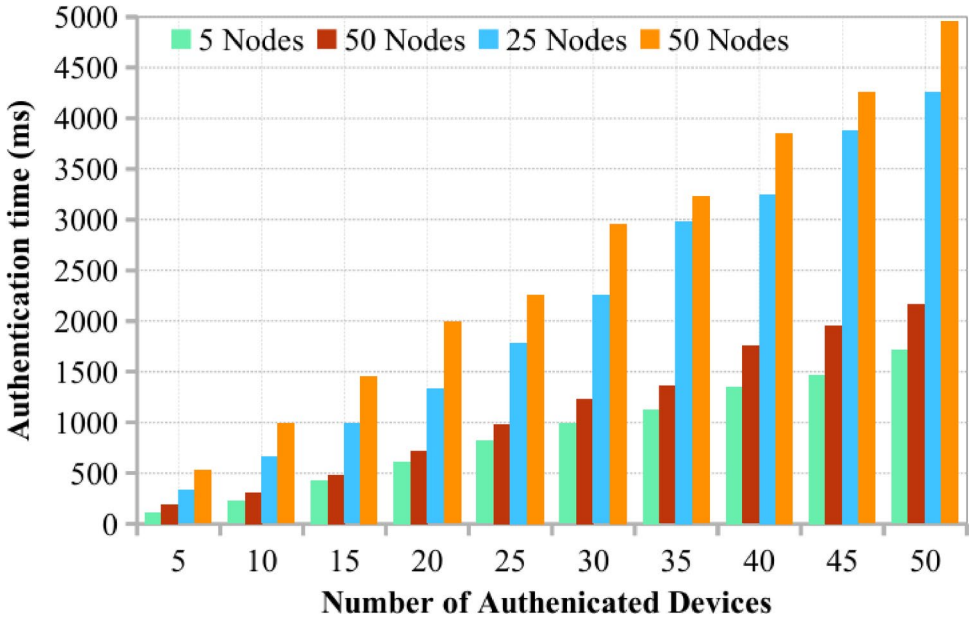

Fig. 4. The authentication time of proposed architecture with different nodes.

designed to scale with the network. This ensures that all nodes, regardless of the network size, are authenticated for secure transmission.

Analysis of the proposed PoAh consensus algorithm

An operational testbed for real-world circumstances and a simulated setting for large-scale research have been used to evaluate the developed PoAh consensus method. This Section presents the specifics of both possibilities, particularly the experiment design used to confirm the sustainability of the suggested model and the analysis of the findings. The PoAh algorithm implements a less resource- and energy-intensive authentication approach. It is beneficial for IIoT designs that are resource constrained. An overview of the critical features of the PoAh consensus algorithm is provided below.

Input: Every node in the blockchain network fllows SHA-256 hash model. Each node has its own private (PrK) and public Keys (PuK)

Output: Block validation of blockchain network.

$B_{Trx+} \rightarrow Blocks$; /* blocks consist of various node transactions

$(Sign_{PrK})(block) \rightarrow Transmissions$; /* Each node transmits blocks with its sign using private key to blockchain network

$(Verify_{PuK})(block) \rightarrow Authentication$; /*Trusted nodes verify block with its public key */

if block validated **then**

$(block)||(PoAh(D) \rightarrow Transmit$; /*Trusted nodes transmit validated block to the blockchain network */

$H(block) \rightarrow Add\ blocks\ to\ the\ blockchain\ network$; /*Mined to the blockchain network */

else

$Drop \rightarrow Remove\ the\ block\ from\ the\ blockchain\ network$; /*Unauthorized node, drop the block */

$(block)||(PoAh(D) \rightarrow Transmit$; /*Trusted nodes transmit validated block to the blockchain network */

return ()

Algorithm. PoAh smart contract.

Resource utilization

An operational testbed for real-world circumstances and a simulated setting for large-scale research have been used to evaluate the developed PoAh consensus method. This Section presents the specifics of both possibilities, particularly the experiment design used to confirm the sustainability of the suggested model and the analysis of the findings. The PoAh algorithm implements a less resource and energy-intensive authentication approach. It is beneficial for resource-constrained IIoT designs. An overview of the critical features of the PoAh consensus algorithm is provided below. For simulation setting we have considered 6 computing nodes among these five nodes are developed to collect data using IoT devices. In our experimentation, we attempted to conduct PoW and PoS on the actual resource-constrained IoT devices; it should be mentioned. We saw that these demanding algorithms could not finish even one mining run. We noticed that even after running the testbed software foundation for days, it still needed to be completed. These powerful consensus methods cannot be implemented on IoT devices with limited resources. Assuming that these IoT devices are battery-operated and placed in potentially remote areas for real-world applications, such as smart cities, it is evident that more than the battery will be required to conduct even one mining operation for these demanding algorithms.

Execution time

Compared to standard PoW, PoAh offers a key advantage of reduced execution time without compromising on security. PoW, which takes about 10 min to validate a single block, is simply not practical for IoT use. In response to this challenge, PoAh employs cryptographic authentication for block validation, a process that is rapid and significantly reduces execution time. This makes PoAh a highly suitable option for IIoT networks with limited resources. The execution time is the total time needed to complete one transaction request authentication, which is defined as follows:

$$\partial\, T_{txr} = T_{ch} - T_i$$

Here T_{ch} represents the specific node timestamp which is added to the blockchain network. T_i is the initial timestamp of node transactions request. The proposed authentication consensus mechanism provides better execution time, which is suitable for IoT applications.

Security

Integrating PoAh as a consensus algorithm into blockchain-based IoT frameworks ensures an IoT framework's sustained security, providing real-time security and accurate data authentication. The security needed for cryptocurrency is different from that required for IoT systems. PoAh, apart from the reverse hash calculation needed for block validation, integrates with the current PoW cryptography paradigm. This integration ensures

that PoAh offers adequate security when combined with ECDSA in decentralized IIoT networks. In addition, PoAh effectively tackles the two main flaws in the existing blockchain consensus: the network's 51% attack and erratic network connection. While only trustworthy nodes can authenticate blocks in PoAh, any network device may produce blocks. This selective notification of peers about adverse circumstances resulting from these shortcomings ensures a more secure system. Blocks can only be added to the chain and authenticated by trusted peers, and the dynamic nature of trust values effectively addresses PoW's vulnerability to a 51% assault. One of the main advantages of the proposed blockchain model is the traceability of unauthorized access. The proposed consensus model based blockchain network transaction T_{txr} can be able to get the random message shared ($PID_{i,j}$) by any node in the network ($N_{i,j}$). If any IoT node is originated to share any illegal message, then the proposed IoT will send back illegal message back to T_{txr}. This will ensure by the public key of the node ($N_{i,j}$), then the real node identity is calculated as follow as:

$$RID_{i,j} = hash(msg.publickey) \oplus PID_{i,j}$$

For the above analysis we can prove that our proposed model can easily trace the origin of any illegal activity.

Performance evaluation of industrial operations

The efficiency study of the proposed blockchain structure, as demonstrated in this section, is crucial for understanding the potential impact of blockchain on industrial procedures. The service time required for various industrial procedures, which was assessed through several trials, is a key metric in determining the efficiency and practicality of the proposed blockchain structure. The duration of a task, comprising run-time or system processes executed on the computer's behalf, is the task's running time or processor time. The five industrial processes carried out by the system at hand are gathering data, movement, information from sensors capturing, device authentication, user authentication, and device identification. In this study, we used a Core i5 desktop computer to conduct thorough simulations in Python to separate consumers, components, documents, and controllers into distinct groups and record the service's execution time. We go over each operation's service duration in the sections that follow.

First, we examine the enrollment procedure for blockchain-enabled intelligent industrial networks, a crucial step in establishing a secure and efficient network. There are five groups of users, with 10, 20, 30, 40, and 50 members each. A single user's registration procedure consists of three primary steps: submitting a transaction request, authenticating the purchase, and updating the blockchain after successful verification. The first group's execution time records a minimum of 815 milliseconds and a high of 1571 milliseconds. For the first group, the median time of execution is 963 ms. The bar graph in Fig. 5 compares the service time required for the various user groups, providing insights into the scalability and performance of the enrollment procedure. The comparison shows that more execution time was needed as the number of users increased. With fifty users, the fifth group is thought to be the largest. This group's average time is 963 milliseconds. The user enrollment procedure responds quickly overall, which is a positive indicator of the efficiency of the proposed blockchain structure.

The sensor's data-capturing procedure is examined in the third stage. Immutable information storage is one of the most essential features of blockchain-enabled technologies. The sensors for the suggested solution are divided into five groups, each with 10, 25, 50, 75, and 100 devices. Sensor identification, data encryption, transactional operation, and sensor data retention based on the device ID are the four steps of the blockchain network's storage of information procedure. Figure 6 displays a bar graph that compares the application's execution times for the various sensor groups used for data recording. The graph indicates that 1908 ms is the mean execution time for

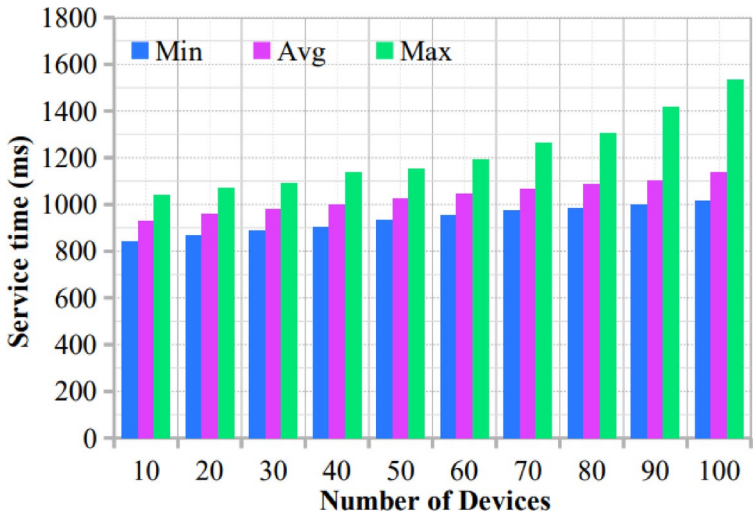

Fig. 5. The service time computation of blockchain devices for user registration.

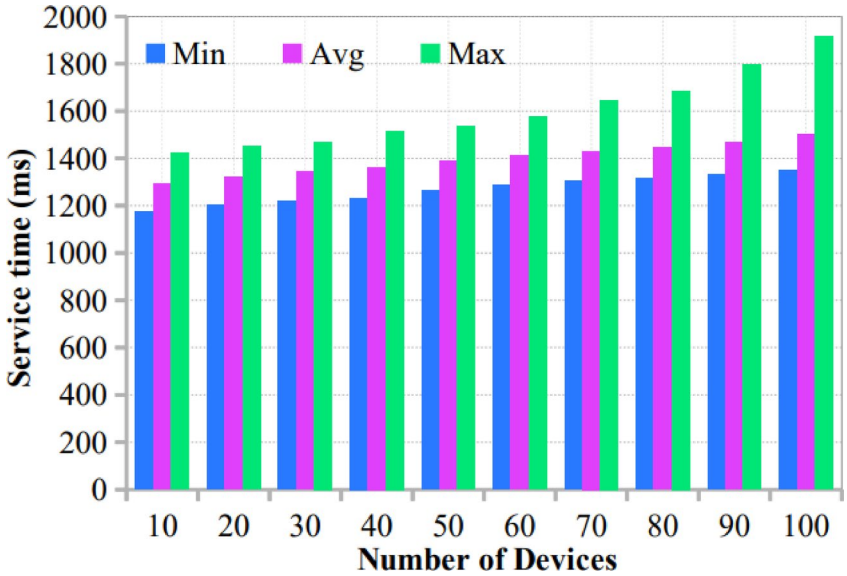

Fig. 6. The service time computation of blockchain devices for task execution.

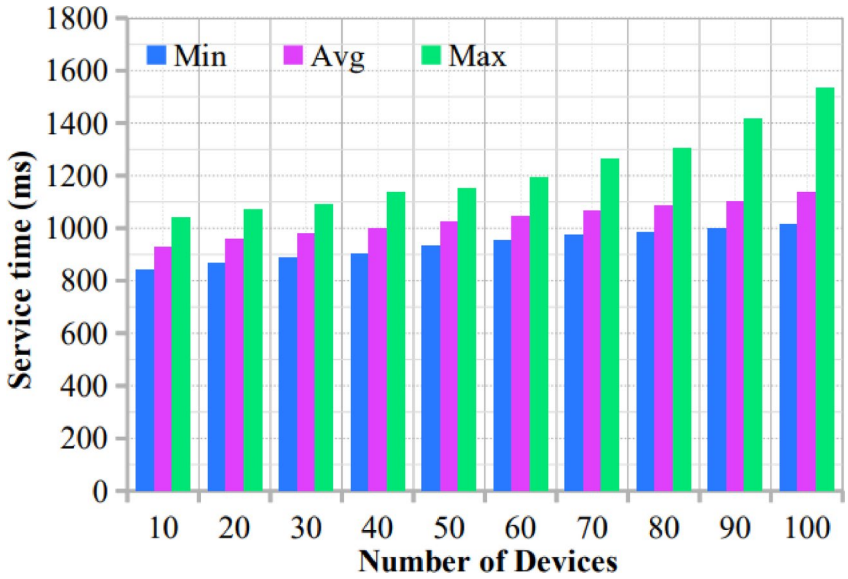

Fig. 7. The service time computation of blockchain devices for storage.

the most extensive set of devices. It shows that the suggested design performs satisfactorily in collecting sensor information.

The authorization of people to view the information collected by the sensor is an additional crucial function. The method consists of three steps:

- Submitting a transaction request with user and device details.
- Authenticating the consumer and device, following effective authentication.
- Granting access to stored data.

The sensor's record is divided into five groups to assess the time required for execution. These groups comprise numerical information about outside characteristics like climate, pressure, size, moisture, etc. A bar graph displaying a comparative examination of the service execution time for the various record groups is depicted in Fig. 7. The graph shows that the procedure was performed more quickly because the total execution time was less than 875 ms. For the most enormous batch of sensor recordings, the average execution time is 1612 ms. It shows that the suggested design performs satisfactorily in recording sensor data. In conclusion, the total service execution time statistics show that the suggested blockchain platform performs satisfactorily for every activity.

C. Blockchain Efficiency.

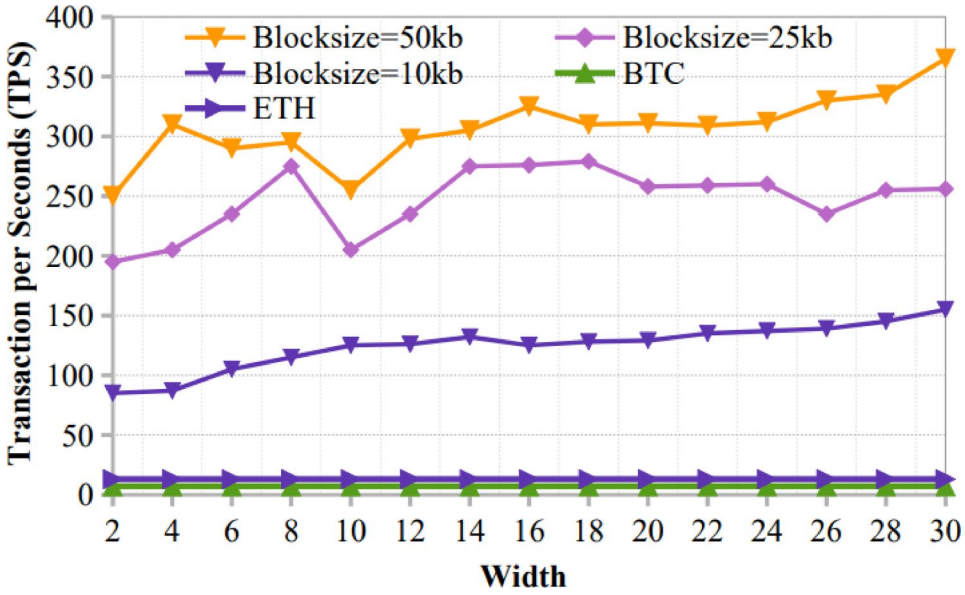

Fig. 8. The proposed architecture block size comparison with baseline model.

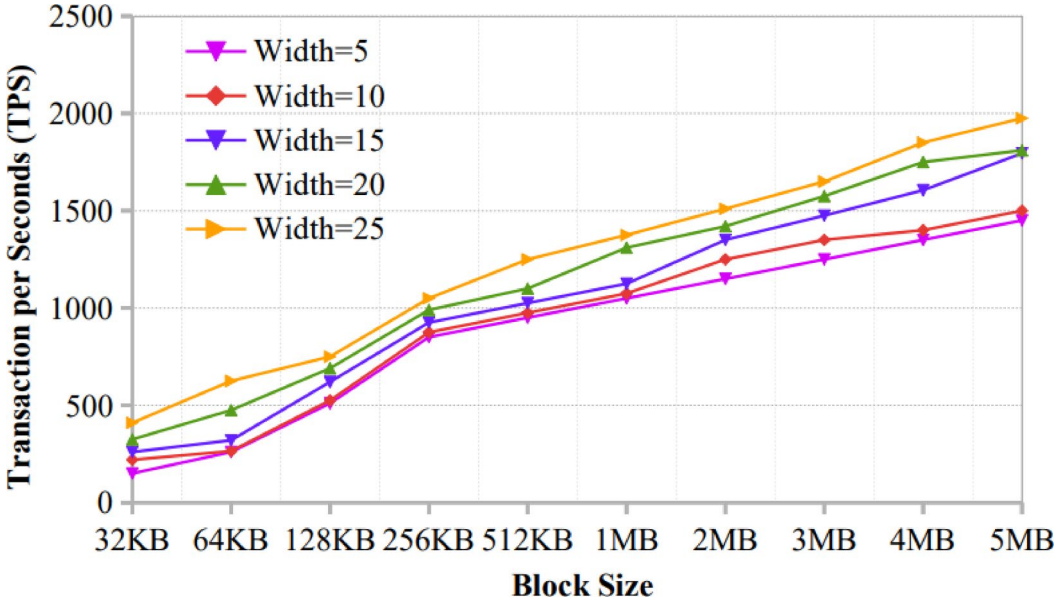

Fig. 9. The proposed architecture width comparison with baseline model.

Our assessment of blockchain's effectiveness in a real-world setting is facilitated by Golang, a key tool in creating the consensus mechanism. This involves the setup of a complex network of three Linux laptops, six PCs, and a cloud server, all intricately interconnected via a local network. Each device functions as a miner, solving complex mathematical puzzles. The Node.js script, a crucial component, generates interactions with randomized locations and amounts. The script is executed by the trigger node, leading to the creation of new transactions that are then deposited in matching pools based on their addresses, awaiting bundling by miners. To simulate a WAN scenario, we manually introduce a propagation delay, with a mean value of 35–55 ms, subject to the typical distribution.

The blockchain's performance is displayed in Figs. 8 and 9 for various block sizes and widths. As the width of the PoAh architecture grows, throughput rises with a fixed block size, as seen in Fig. 8. Meanwhile, throughput increases proportionally to block size growth, assuming a fixed width. Our blockchain achieves 225 tps, a significant improvement over Bitcoin and Ethereum, which achieves 7 tps and 27 tps respectively. The link between throughput in various block sizes and widths is also shown on Fig. 9. It is evident that the greater width and block size values of the proposed blockchain behave better than the more compact one. It reaches its

maximum throughput of 1273 tps, a substantial improvement over Bitcoin and Ethereum, where block size is 2 MB and width is 15.

Our experiments showed that the blockchain-assisted edge computing architecture significantly reduced latency and improved service time compared to conventional IIoT systems. The inclusion of edge nodes, which localized data processing using blockchain, was the main reason for the decreased latency. The blockchain layer, with its trust mechanism, ensured data integrity and security for transactions between IoT devices and edge nodes. Importantly, the system successfully repelled simulated security risks, including replay and man-in-the-middle attacks, demonstrating the robustness of the system. The proposed system performs well with a moderate number of IoT devices; however, performance decreased when the blockchain network's participating edge nodes grew above 100. This bottleneck is due to the increased challenge of reaching a consensus with more nodes.

Conclusion

Industrial IoT modernizes innovative sectors by integrating cutting-edge technologies. One such technology that has drawn significant interest from academia and industry is blockchain. An intelligent industry may benefit from the advanced characteristics that blockchain technology brings, such as higher efficiency, protection, consistency, trust, and decentralization. In this study, we provide a blockchain-based framework for intelligent edge computing. A safe, lightweight, decentralized private blockchain-based IIoT network that can perform several crucial tasks, including trusted machine management, data storage, and user and device registration, is made possible by the PoAh-based blockchain system. Specifically, we present an edge-level implementation of authentication that lowers the computational burden and improves the industry platform's security. Many performance characteristics are considered while evaluating the suggested architecture's efficiency. The experimental findings demonstrated that the proposed design outperformed other cutting-edge approaches. Ultimately, we used the suggested design to convert a conventional fruit processing facility to realize the enormous potential of blockchain technology for an intelligent industry sector.

Data availability

This manuscript does not report data generation or analysis.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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